

Is Gen AI Bad For The Environment?

Posted by u/JCunliffeUK • 0 points • 43 comments

Been enjoying making comics like the rest of the community, but saw a couple of comments saying that Gen AI is bad for the environment. I asked ChatGPT to do some deep research on the topic and give me some comparable items that most people use, so I had a better visual understanding of Gen AI's carbon footprint.

I was surprised at how little it actually was.

Am I missing anything, or is the sustainability argument against the use of AI a weak one?

Comments

u/Jeff_NZ • 1 points • 2025-04-17 00:46 UTC

Also read Grok 3 paper on AGW as another perspective on actual impact.

u/TheJzuken • 4 points • 2025-04-16 13:27 UTC

Generating images with 4o is probably closer to charging a smartphone or even 3D rendering for 5 minutes (that's how long it takes to generate a 4o image), but it's still very small compared to the whole economy.

Even by 2030 data centers will consume less than other industries:

[<https://www.datacenterdynamics.com/en/news/iea-global-data-center-electricity-consumption-to-increase-significantly-but-remain-a-small-part-of-overall-usage/>]

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[https://preview.redd.it/07z2oe0187ve1.png?](https://preview.redd.it/07z2oe0187ve1.png?width=1198&format=png&auto=webp&s=4940e3ee254f8d8e7512666b02a7bc7a01cab820)

[width=1198&format=png&auto=webp&s=4940e3ee254f8d8e7512666b02a7bc7a01cab820](https://preview.redd.it/07z2oe0187ve1.png?width=1198&format=png&auto=webp&s=4940e3ee254f8d8e7512666b02a7bc7a01cab820)

u/BrightSkyFire • -3 points • 2025-04-16 12:56 UTC

Nothing like a “discussion post” where the OP is literally using AI for every step of the discussion from the comic, to the “sources”, and I’m fairly sure they’re even using ChatGPT to write their comment responses.

There are people in this world who would literally rather die than do any of their own thinking. Shit is insane.

u/JCunliffeUK • 4 points • 2025-04-16 13:44 UTC

Care to find some contradictory sources so we can all learn together? Or is your intention to have an emotional outburst and be rude when presented with facts?

Because if there's better information out there, I would actually love to read it so I can be better informed. I am an open-minded person who is willing to learn, you just need to provide that information so I can do that. That's the point of a discussion thread.

Honestly, what's going on with the quality of reddit when ad hominem is the norm across the comment section rather than discussing the actual topic of the post.

u/BrightSkyFire • -3 points • 2025-04-16 14:31 UTC

> Because if there's better information out there, I would actually love to read it so I can be better informed. I am an open-minded person who is willing to learn, you just need to provide that information so I can do that. That's the point of a discussion thread.

If that were true, you would go to any number of publically available scientific journal websites and found any number of papers outlining the hefty costs associated with image generation. Instead, you asked ChatGPT to do all the hard work for you.

You aren't interested in a genuine answer. Your lack of independent research beyond ChatGPT's provided sources speaks far louder than your ChatGPT approved messages insisting you're interested in discussion.

u/JCunliffeUK • 5 points • 2025-04-16 15:01 UTC

I'm embarrassed by the lack of a response in this response. At least provide some value by sharing an accredited source that you believe in so I can explore the topic a little more.

I could Google 50 websites and still not find my way to your opinion, dude. Like, help me understand like you do. That's why you're here right? To contribute, even with you being a bit rude I'm still willing to listen to gain a deeper understanding on the topic.

And that's a rare thing, I think a lot of people would have just told you to do one, but you're so opinionated regarding this topic that I actually want to hear what you have to say, in case you know something that I don't.

So sincerely, what's your opinion of generative AI and the environment?

u/More-Ad5919 • 1 points • 2025-04-16 12:18 UTC

And what did this comparison tell us now? Seems really bad to me.

u/JCunliffeUK • 1 points • 2025-04-16 12:25 UTC

It's supposed to make us think about it and better understand the ecological impact in comparison to the other things we do in life.

u/More-Ad5919 • 1 points • 2025-04-16 12:37 UTC

Good.

u/Resident-Mine-4987 • 1 points • 2025-04-16 12:16 UTC

Ahhh yes. I love the argument of "if it's not the worst then it must be good"

u/JCunliffeUK • 2 points • 2025-04-16 12:20 UTC

Where do you think the line is? That's what I'm trying to understand

u/sanftewolke • -3 points • 2025-04-16 12:04 UTC

An AI comic about the small environmental impact of AI for by an OP who is commenting using only AI. What kind of a conversation is that?

And to OP: if you are an actual human and not just some bot, maybe try to engage in actual human interaction instead of trashing the internet with AI comments.

AI is great for many things. Not for putting an OpenAI filter on human communication

u/JCunliffeUK • 4 points • 2025-04-16 12:16 UTC

Bit mean, I'm assuming you're talking about the sources and Deep Research conversation which were in response to a user that was asking for them. I can assure you that I'm a real person who has interpreted the data to put together the comic.

It seems like a lot of people have been offended by this and I can't figure out why. Is the information not useful to better understand generative AI's ecological impact?

It seems like Deep Research is kind of like doing a Google Search but it compiles information for multiple websites. In this case over 20.

I don't know, the tone of... Don't use AI tools to source information about AI topics seems irrelevant when it's sourcing from websites that aren't AI websites.

Now, do you actually have thoughts on the environmental aspects and topic of gen AI or do you want to throw another insult at me?

u/sanftewolke • 0 points • 2025-04-16 13:05 UTC

Yeah was a bit mean, I apologize for that. But even in your response you sound like you format your answers with Chatgpt.

In this answer about answering with Ai you sound like AI.

If you someone sounds like their answers could be featured in "I, robot" then, yeah, I assume it's AI

u/JCunliffeUK • 3 points • 2025-04-16 13:27 UTC

Robot is a common insult thrown at people on the spectrum, I guess. So even in your ad hominem, you're not being original.

It's common for people who are emotionally triggered to not engage in the subject of debate and instead attack someone's character. So I'm sorry if the contents of this post has upset you, it wasn't my intention to trigger you. It was only made to educate and inform.

u/sanftewolke • 2 points • 2025-04-16 13:49 UTC

Well if that's the case, I truly apologize. If someone makes a post about how a downside of AI is actually not bad, is obviously using AI for some answers and then answering every comment with quite similar suspiciously AI-sounding comments I think to assume AI is not too far off.

But I might be wrong, could have all been directed against the wrong person

u/JCunliffeUK • 2 points • 2025-04-16 13:56 UTC

Hey, I get it, this subject matter isn't easy. I've been on both sides of it, and it'd be weirder if people didn't have strong opinions. I'm just seeking the truth on it all. I appreciate the apology.

u/Free-Design-9901 • -3 points • 2025-04-16 11:19 UTC

This comparison looks bad for AI images. So using a smartphone equals to generating 4 AI images?

Smartphone is the most useful tool invented by man.

Generating AI images is something people do when they're bored of TikTok. On their smartphone.

u/JCunliffeUK • 4 points • 2025-04-16 11:28 UTC

It's valid for you to feel like smartphones have more value than images.

The question this poses though is, do you feel 226x as bad when you stream video for an hour? or 70-166x as bad if you play a video game for an hour? or 40x worse if you use a microwave?

This was the first time I'm seeing the numbers and thinking about this stuff. So I'm trying to understand how I should feel about it, considering my own use of gen AI. After this, I don't feel like I'm causing that much ecological harm in the grand scheme of things.

But I'm also trying to understand the wider energy costs of training models and the costs of data centers. At the end of the day, I just want honest data so I can stay informed.

u/FlanSteakSasquatch • 2 points • 2025-04-16 11:11 UTC

Creating a single gen AI image is not a significant amount of power. But we are creating a lot, and doing a lot more than just creating images.

You can find plenty of info with a quick google search, but [power consumption of data centers are huge](https://www.rand.org/pubs/research_reports/RR43572-1.html). Individually they tend to use about as much power as a small city. That article estimates the total world AI data centers will match the power consumption of the entire state of California by 2027. The carbon footprint of gen ai is definitely large.

u/ForceTypical • 3 points • 2025-04-16 11:23 UTC

“As much as a small city” is ridiculous and completely untrue. The ai data centre industry uses something like 1/100th of the power that the steel industry uses. You don't seem to have a problem with that though. Not to mention most ai companies keep their data centres hidden and private, so most so called “data” is purely speculative and usually an overestimation. Keep in mind these companies are businesses, and will try to save money wherever possible. If they find they are using that much power, they will try their best to reduce the amount of power they are using to save money, because that would be INSANELY expensive.

u/FlanSteakSasquatch • 1 points • 2025-04-16 12:11 UTC

I mean I totally agree that the steel industry is using a huge amount of power, even orders of magnitude more than compute, not sure why you said I “don't seem to have a problem with that”.

But the statement about data centers using as much power as a small city is verifiably true. We are building gigawatt+ data centers and there are megawatt cities. So we will have data centers that match the lower consumption of a few hundred small cities or more.

u/JCunliffeUK • 2 points • 2025-04-16 12:48 UTC

Can I just say thank you to both of you for actually discussing the topic of the post, I really do want to understand the data center angle more. I appreciate you both contributing.